

```
1  public class Stack<T> {
2
3      private Node<T> head;
4      public Stack() {
5          this.head = null;
6      }
7
8      public boolean isEmpty() {
9          return this.head == null;
10     }
11
12     // Inserts value x to the head of the stack
13     public void push(T x) {
14         this.head = new Node<T>(x, this.head);
15     }
16
17     // Removes and returns the value at the head
18     public T pop() {
19         T x = this.head.getValue();
20         this.head = this.head.getNext();
21         return x;
22     }
23
24     // Returns the value at the head without removing it
25     public T top() {
26         return this.head.getValue();
27     }
28
29     // Returns a description of the stack as [x1, x2, ..., xn]
30     public String toString() {
31         if (this.isEmpty()) {return "["; }
32
33         String s = "[";
34         Node<T> temp = this.head;
35         while (temp != null) {
36             s += temp.getValue().toString();
37             if (temp.hasNext()) {
38                 s += ", ";
39             }
40             temp = temp.getNext();
41         }
42
43         s += "]";
44         return s;
45     }
46 }
```